

NeetCode 250 — Beginner Roadmap

Prerequisites · All 18 Phases · ~160 Days

Prerequisites (Complete Before Touching LeetCode)

Spend 1 week making sure you are comfortable with all of the following. If any of these feel unfamiliar, practice them in isolation first.

Topic	What You Need	Quick Check
Python Data Structures	list, dict, set — creation, indexing, slicing, iteration, membership test (in)	If you cannot write a for loop over a dict.items() confidently, stop here first
Loops & Conditionals	for, while, nested loops, break/continue, list comprehensions	Practice: print all pairs (i,j) where i+j = 10 from a list of numbers
Functions & Recursion	def, return, default args, *args. Recursion: factorial, fibonacci from scratch	If recursion feels like magic, write fibonacci both iteratively and recursively
Built-in Collections	collections.Counter, collections.defaultdict, collections.deque	Know: Counter("hello"), defaultdict(int), deque with appendleft/popleft
Sorting	sorted(), list.sort(), key=lambda, reverse=True. Sort list of tuples by second element	nums.sort(key=lambda x: x[1]) — you will use this constantly
heapq Module	heapq.heappush, heapq.heappop, heapq.nlargest, heapq.nsmallest, heapify	Python heapq is a min-heap by default. Negate values for max-heap behavior
String Operations	split, join, strip, replace, s[::-1], ord/chr, isalpha, isdigit, lower/upper	Know how to check if a char is alphanumeric without importing anything
Sets and Hashmaps	set operations: add, discard, union, intersection. dict: get, items, pop, setdefault	Understand why dict lookup is O(1) average — the hash function
Big-O Notation	Understand O(1), O(log n), O(n), O(n log n), O(n ²), O(2 ⁿ). Know which is better	Before submitting any solution, always ask: what is my time and space complexity?
Two-pointer Mental Model	Understand the idea of left/right indices moving toward each other on an array	Practice on paper: find if sorted array [1,2,3,6,8] has pair summing to 9
Recursion Stack Model	Be able to trace what happens when a recursive function calls itself	Draw the call tree for fibonacci(4) on paper before starting tree problems
Class Basics	class definition, __init__, self, creating objects — needed for Linked List nodes	class ListNode: def __init__(self, val=0, next=None): self.val=val; self.next=next

Timeline: ~160 days at 1–2 problems/day · 90–100 days at 2–3 problems/day · 18 phases · 250 problems

Phase 1 — Arrays & Hashing

Days 8–17 · 22 problems

Pattern: *O(1)* lookup · frequency counting · prefix sums · hashmap complement

Week 1

LC#	Problem	Diff	Key Technique
1929	Concatenation of Array	Easy	Warmup — basic array ops and indexing
217	Contains Duplicate	Easy	Hashset existence check, $O(n)$ vs $O(n^2)$
242	Valid Anagram	Easy	Counter comparison or sort both strings
1	Two Sum	Easy	Complement in hashmap — most asked ever
169	Majority Element	Easy	Boyer-Moore voting algorithm
383	Ransom Note	Easy	Counter subtraction check
953	Verifying an Alien Dictionary	Easy	Custom order hashmap lookup

Week 2

LC#	Problem	Diff	Key Technique
36	Valid Sudoku	Med	Hashset per row, col, and 3x3 box
49	Group Anagrams	Med	Sorted tuple as dictionary key
347	Top K Frequent Elements	Med	Counter + heap or bucket sort
238	Product of Array Except Self	Med	Prefix array $\times$ suffix array, no division
271	Encode and Decode Strings	Med	Length-prefix encoding scheme
560	Subarray Sum Equals K	Med	Prefix sum + frequency hashmap
525	Contiguous Array	Med	Prefix sum with 0 converted to -1
128	Longest Consecutive Sequence	Med	Hashset + streak building from start
229	Majority Element II	Med	Boyer-Moore extended for $n/3$ threshold
974	Subarray Sums Divisible by K	Med	Prefix sum mod k in hashmap
554	Brick Wall	Med	Hashmap of gap edge positions
2352	Equal Row and Column Pairs	Med	Serialize rows into hashmap, match cols

Phase 2 — Two Pointers

Days 18–24 · 13 problems

Pattern: Sorted pair-finding · palindrome · in-place removal · squeeze from both ends

Week 3

LC#	Problem	Diff	Key Technique
125	Valid Palindrome	Easy	l/r converge, skip non-alphanumeric
27	Remove Element	Easy	Slow/fast overwrite with valid values
26	Remove Duplicates from Sorted Array	Easy	Slow/fast, keep unique values only
283	Move Zeroes	Easy	Two-ptr partition zeros to back
167	Two Sum II (Sorted Input)	Med	Classic opposite-end on sorted array
75	Sort Colors	Med	Dutch National Flag 3-way partition
881	Boats to Save People	Med	Greedy two-ptr on sorted array
15	3Sum	Med	Fix i, two-ptr rest, skip duplicates
11	Container With Most Water	Med	Greedy: move shorter side inward
18	4Sum	Med	Fix 2 pointers + two-ptr remaining
2095	Delete the Middle Node	Med	Fast/slow pointers to find middle
80	Remove Duplicates Sorted Array II	Med	Allow up to 2 occurrences
42	Trapping Rain Water	Hard	Precompute left/right max or two-ptr

Phase 3 — Sliding Window

Days 25–31 · 9 problems

Pattern: Contiguous subarray/substring · expand right shrink left · variable window constraint

Week 4

LC#	Problem	Diff	Key Technique
121	Best Time to Buy and Sell Stock	Easy	Track min so far, max profit
1004	Max Consecutive Ones III	Med	Flip up to k zeros window
1493	Longest Subarray of 1s After Deleting	Med	At most one zero in window
3	Longest Substring Without Repeating	Med	Hashset tracks window chars
567	Permutation in String	Med	Fixed-size window, Counter compare
424	Longest Repeating Character Replacement	Med	Freq map + max_freq trick
2461	Max Sum of Distinct Subarrays of Size K	Med	Fixed-size + distinct element check
76	Minimum Window Substring	Hard	have/need counters, shrink loop
239	Sliding Window Maximum	Hard	Monotonic deque maintains max

Phase 4 — Stack

Days 32–39 · 14 problems

Pattern: LIFO bracket matching · expression eval · monotonic stack for next greater/smaller

Week 5

LC#	Problem	Diff	Key Technique
20	Valid Parentheses	Easy	Push open, pop+match close brackets
155	Min Stack	Med	Store pairs (val, current_min)
150	Evaluate Reverse Polish Notation	Med	Push nums, pop 2 on each operator
71	Simplify Path	Med	Stack, split on slash characters
394	Decode String	Med	Stack handles nested repeat groups
735	Asteroid Collision	Med	Stack simulates collisions step by step
496	Next Greater Element I	Easy	Monotonic stack + result hashmap

Week 6

LC#	Problem	Diff	Key Technique
739	Daily Temperatures	Med	Decreasing mono-stack stores indices
503	Next Greater Element II Circular	Med	Double the array trick for wrap
853	Car Fleet	Med	Sort by position, stack arrival times
901	Online Stock Span	Med	Mono-stack of (price, span) pairs
84	Largest Rectangle in Histogram	Hard	Increasing stack, calc area on pop
85	Maximal Rectangle	Hard	Histogram DP applied row by row
22	Generate Parentheses	Med	Skip — revisit in Backtracking phase

Phase 5 — Binary Search

Days 40–47 · 14 problems

Pattern: Sorted/rotated search · boundary finding · binary search on answer space

Week 7

LC#	Problem	Diff	Key Technique
704	Binary Search	Easy	Write template from scratch 3 times
74	Search a 2D Matrix	Med	Mentally flatten or row-then-col
34	Find First and Last Position	Med	Two separate binary searches
162	Find Peak Element	Med	Compare mid to mid+1
153	Find Min in Rotated Sorted Array	Med	Compare mid to right half
33	Search in Rotated Sorted Array	Med	Identify which half is sorted
981	Time Based Key-Value Store	Med	Binary search on timestamp list

Week 8

LC#	Problem	Diff	Key Technique
875	Koko Eating Bananas	Med	First answer-space binary search
1011	Capacity to Ship Packages	Med	Same answer-space pattern as Koko
658	Find K Closest Elements	Med	Binary search on window start index
240	Search a 2D Matrix II	Med	Start top-right, eliminate row or col
2300	Successful Pairs of Spells	Med	Binary search per query spell
2560	House Robber IV	Med	Binary search on answer + greedy check
4	Median of Two Sorted Arrays	Hard	Binary search on partition — hardest

Phase 6 — Linked List

Days 48–55 · 14 problems

Pattern: Pointer manipulation · dummy head · fast/slow pointers · Floyd cycle detection

Week 9

LC#	Problem	Diff	Key Technique
206	Reverse Linked List	Easy	prev/curr/next iterative — draw first
21	Merge Two Sorted Lists	Easy	Dummy head merge pointer
141	Linked List Cycle	Easy	Fast/slow meet = cycle exists
2095	Delete the Middle Node	Med	Fast/slow find middle node
19	Remove Nth Node from End	Med	Two pointers n apart
92	Reverse Linked List II	Med	Reverse sublist with prev anchor
82	Remove Duplicates II	Med	Dummy head + skip duplicate runs

Week 10

LC#	Problem	Diff	Key Technique
2	Add Two Numbers	Med	Carry simulation with dummy head
138	Copy List with Random Pointer	Med	Hashmap old node to new node
287	Find the Duplicate Number	Med	Floyd's cycle detection on array
2130	Maximum Twin Sum of Linked List	Med	Find mid, reverse 2nd half, sum
146	LRU Cache	Med	Doubly-linked list + hashmap design
25	Reverse Nodes in k-Group	Hard	Group-by-group reversal
23	Merge K Sorted Lists	Hard	Min-heap of (val, node) — revisit later

Phase 7 — Trees

Days 56–70 · 23 problems

Pattern: Recursive tree thinking · post-order returns upward · BFS levels · BST property

Week 11 — Basic Recursion

LC#	Problem	Diff	Key Technique
226	Invert Binary Tree	Easy	Swap children recursively
104	Maximum Depth of Binary Tree	Easy	1 + max(left depth, right depth)
100	Same Tree	Easy	Compare node by node recursively
572	Subtree of Another Tree	Easy	isSameTree called at every node
110	Balanced Binary Tree	Easy	Height = -1 signals imbalance upward
112	Path Sum	Easy	DFS subtract target as you descend
543	Diameter of Binary Tree	Easy	Post-order track global max separately

Week 12 — BFS + BST

LC#	Problem	Diff	Key Technique
102	Binary Tree Level Order Traversal	Med	BFS queue, collect nodes by level
199	Right Side View	Med	BFS last node each level
98	Validate BST	Med	Pass (lo, hi) bounds down recursively
230	Kth Smallest in BST	Med	Inorder traversal = sorted sequence
235	LCA of BST	Med	Use BST property to navigate left/right
450	Delete Node in BST	Med	Find inorder successor or predecessor
669	Trim a BST	Med	Recursively prune out-of-range nodes
1038	BST to Greater Sum Tree	Med	Reverse inorder with running sum

Week 13 — Harder Trees

LC#	Problem	Diff	Key Technique
1448	Count Good Nodes in Binary Tree	Med	DFS pass max-so-far down
113	Path Sum II	Med	DFS backtracking collect full path
236	LCA of Binary Tree	Med	Return node if found, merge results
105	Construct from Preorder+Inorder	Med	Root index splits inorder array
116	Populating Next Right Pointers	Med	BFS link nodes level by level
437	Path Sum III	Med	Prefix sum hashmap on DFS path
124	Binary Tree Maximum Path Sum	Hard	Post-order global max tracking
297	Serialize and Deserialize Tree	Hard	Preorder with null markers

Phase 8 — Backtracking

Days 71–79 · 17 problems

Pattern: Choose → Explore → Unchoose · sort for pruning · use start index for combinations

Week 14

LC#	Problem	Diff	Key Technique
78	Subsets	Med	Include/exclude each element
77	Combinations	Med	C(n,k) backtrack with start index
17	Letter Combinations Phone Number	Med	Map digits, chain character choices
46	Permutations	Med	Swap-based or used[] boolean array
90	Subsets II (with duplicates)	Med	Sort + skip same value at same level
40	Combination Sum II	Med	Sort + skip duplicate branches
47	Permutations II	Med	Sort + skip adjacent duplicates
39	Combination Sum	Med	Reuse allowed: use i not i+1

Week 15

LC#	Problem	Diff	Key Technique
131	Palindrome Partitioning	Med	Backtrack + palindrome check helper
93	Restore IP Addresses	Med	4-part backtrack with range validation
22	Generate Parentheses	Med	Track open/close count constraints
79	Word Search	Med	DFS + in-place visited marking
1239	Max Length of Concatenated Chars	Med	Backtrack unique-char bitmask strings
51	N-Queens	Hard	Row-by-row, check col/diagonal sets
52	N-Queens II	Hard	Count solutions only — same structure
37	Sudoku Solver	Hard	Backtrack fill empty cells row by row
212	Word Search II	Hard	Trie + backtracking — do after Tries

Phase 9 — Heap / Priority Queue

Days 80–86 · 12 problems

Pattern: Min-heap default · negate for max-heap · two heaps for median · K-element problems

Week 16

LC#	Problem	Diff	Key Technique
1046	Last Stone Weight	Easy	Max-heap simulate collisions
703	Kth Largest Element in a Stream	Easy	Min-heap maintained at size k
973	K Closest Points to Origin	Med	Max-heap of size k on distance
215	Kth Largest Element in Array	Med	Min-heap or quickselect
347	Top K Frequent Elements	Med	Revisit: Counter + heap
767	Reorganize String	Med	Max-heap greedy interleave chars
621	Task Scheduler	Med	Greedy + max-heap + cooldown
1845	Seat Reservation Manager	Med	Min-heap of free seat numbers
373	Find K Pairs with Smallest Sums	Med	Min-heap of pairs expansion
378	Kth Smallest in Sorted Matrix	Med	Min-heap row-by-row expansion
23	Merge K Sorted Lists	Hard	Min-heap of (val, node) — revisit
295	Find Median from Data Stream	Hard	Two heaps: max-left, min-right

Phase 10 — Tries

Days 87–90 · 4 problems

Pattern: TrieNode with children dict + end flag · $O(L)$ insert and search · prefix matching

Week 17

LC#	Problem	Diff	Key Technique
208	Implement Trie (Prefix Tree)	Med	TrieNode children dict + end_of_word
211	Design Add and Search Words	Med	Trie + DFS for dot wildcard
1268	Search Suggestions System	Med	Trie or sort + binary search
212	Word Search II	Hard	Trie + backtracking grid DFS

Phase 11 — Graphs

Days 91–104 · 21 problems

Pattern: BFS/DFS on adjacency list or grid · visited set · Union-Find · topological sort

Week 18 — Grid BFS/DFS

LC#	Problem	Diff	Key Technique
200	Number of Islands	Med	DFS/BFS mark visited cells as 0
695	Max Area of Island	Med	DFS return area count recursively
994	Rotting Oranges	Med	Multi-source BFS from all rotten
286	Walls and Gates	Med	Multi-source BFS from all gates
130	Surrounded Regions	Med	DFS from border, mark safe cells
417	Pacific Atlantic Water Flow	Med	Reverse BFS from both ocean borders
133	Clone Graph	Med	DFS + hashmap old node to new node

Week 19 — Union-Find + Topo Sort

LC#	Problem	Diff	Key Technique
547	Number of Provinces	Med	DFS / BFS / Union-Find components
684	Redundant Connection	Med	Union-Find detect first cycle edge
323	Number of Connected Components	Med	Union-Find or DFS count groups
261	Graph Valid Tree	Med	n-1 edges + no cycle = valid tree
207	Course Schedule	Med	DFS cycle detection white/grey/black
210	Course Schedule II	Med	DFS topological sort post-order

Week 20 — Weighted Graphs

LC#	Problem	Diff	Key Technique
1091	Shortest Path in Binary Matrix	Med	BFS 8-directional grid traversal
127	Word Ladder	Hard	BFS on word neighbors graph
743	Network Delay Time	Med	Dijkstra from source with min-heap
787	Cheapest Flights within K Stops	Med	Bellman-Ford k iterations
1514	Path with Maximum Probability	Med	Dijkstra max-heap variant
332	Reconstruct Itinerary	Hard	Hierholzer's Eulerian path
269	Alien Dictionary	Hard	Topological sort on char dependencies
663	Number of Provinces (variant)	Med	Union-Find path compression

Phase 12 — 1D Dynamic Programming

Days 105–118 · 17 problems

Pattern: $dp[i]$ = optimal subproblem i · find recurrence · fill bottom-up · $O(1)$ space trick

Week 21 — Foundation DP

LC#	Problem	Diff	Key Technique
70	Climbing Stairs	Easy	$dp[i] = dp[i-1] + dp[i-2]$ — feel recurrence
746	Min Cost Climbing Stairs	Easy	$dp[i] = cost + \min(dp[i-1], dp[i-2])$
198	House Robber	Med	Take or skip, track prev2 and prev1
213	House Robber II	Med	Run twice: $[0, n-1]$ and $[1, n]$
53	Maximum Subarray	Med	Kadane's: $cur = \max(num, cur + num)$
152	Maximum Product Subarray	Med	Track min and max simultaneously
322	Coin Change	Med	Unbounded knapsack bottom-up fill

Week 22 — String DP

LC#	Problem	Diff	Key Technique
5	Longest Palindromic Substring	Med	Expand from each center position
647	Palindromic Substrings	Med	Expand from center, count valid ones
91	Decode Ways	Med	$dp[i]$ depends on 1 or 2 prev digits
139	Word Break	Med	$dp[i] = \text{any } dp[j] \text{ where } s[j:i] \text{ in dict}$
300	Longest Increasing Subsequence	Med	Patience sort or $O(n^2)$ DP
416	Partition Equal Subset Sum	Med	0/1 knapsack subset-sum variant
494	Target Sum	Med	Count ways to reach target value
1049	Last Stone Weight II	Med	Partition sum DP variant
2140	Solving Questions With Brainpower	Med	DP right to left from end
790	Domino and Tromino Tiling	Med	Recurrence with 3 distinct states

Phase 13 — Greedy

Days 119–125 · 14 problems

Pattern: Locally optimal choice = globally optimal · sort first · scan once · prove no backtrack

Week 23

LC#	Problem	Diff	Key Technique
409	Longest Palindrome	Easy	Count pairs, add 1 if any odd char
55	Jump Game	Med	Track max reachable index as you go
45	Jump Game II	Med	BFS-level greedy minimize jumps
134	Gas Station	Med	Total $\geq$ 0 means solution exists, find start
763	Partition Labels	Med	Last occurrence map, extend range greedily
678	Valid Parenthesis String	Med	Track lo/hi range of open bracket counts
846	Hand of Straights	Med	Counter + greedy from smallest key
1899	Merge Triplets to Form Target	Med	Greedy eliminate invalid triplets
11	Container With Most Water	Med	Revisit with greedy framing
135	Candy	Hard	Two-pass left-right greedy
857	Minimum Cost to Hire Workers	Hard	Sort by ratio, heap for min cost
502	IPO	Hard	Two heaps: locked and available
2402	Meeting Rooms III	Hard	Two heaps: rooms and ongoing meetings
53	Maximum Subarray	Med	Revisit — now frame explicitly as greedy

Phase 14 — Intervals

Days 126–130 · 7 problems

Pattern: Sort by start · compare current.start vs prev.end · min-heap for room counting

Week 24

LC#	Problem	Diff	Key Technique
252	Meeting Rooms	Easy	Sort by start, check consecutive overlap
56	Merge Intervals	Med	Sort + extend end if overlap found
57	Insert Interval	Med	Find overlap range, merge into one
435	Non-overlapping Intervals	Med	Greedy keep interval with earliest end
253	Meeting Rooms II	Med	Min-heap tracks earliest room end
452	Min Arrows to Burst Balloons	Med	Greedy shoot at end of first balloon
1851	Min Interval to Include Query	Hard	Min-heap sorted by interval size

Phase 15 — 2D Dynamic Programming

Days 131–143 · 16 problems

Pattern: $dp[i][j]$ depends on neighbors · grid paths · string comparison · knapsack 2D

Week 25 — Grid DP

LC#	Problem	Diff	Key Technique
62	Unique Paths	Med	$dp[i][j] = dp[i-1][j] + dp[i][j-1]$
64	Minimum Path Sum	Med	Same recurrence, minimize total
221	Maximal Square	Med	$dp[i][j] = \min(\text{top}, \text{left}, \text{diagonal}) + 1$
309	Best Time Buy/Sell with Cooldown	Med	3-state DP: held / sold / rest
518	Coin Change II	Med	Unbounded knapsack 2D variant
97	Interleaving String	Med	2D DP on both string positions
494	Target Sum	Med	Revisit as 2D subset-sum count
1312	Minimum Insertion Steps	Med	LCS-based palindrome DP

Week 26 — String DP + Hard

LC#	Problem	Diff	Key Technique
1143	Longest Common Subsequence	Med	Match or take max of skip — core
72	Edit Distance	Med	Insert/delete/replace recurrence
115	Distinct Subsequences	Hard	Count ways to form target string
44	Wildcard Matching	Hard	* rules similar to regex DP
10	Regular Expression Matching	Hard	2D DP with * wildcard edge cases
329	Longest Increasing Path in Matrix	Hard	DFS + memoization on grid
312	Burst Balloons	Hard	Interval DP, last burst in range
174	Dungeon Game	Hard	DP right-to-bottom minimum HP

Phase 16 — Advanced Graphs

Days 134–140 · 10 problems

Pattern: Dijkstra · Bellman-Ford · Kruskal MST · Tarjan bridges · BFS with bitmask state

Week 27

LC#	Problem	Diff	Key Technique
743	Network Delay Time	Med	Dijkstra min-heap from source
1631	Path with Minimum Effort	Med	Dijkstra minimize max edge on path
778	Swim in Rising Water	Hard	Dijkstra or binary search + BFS
787	Cheapest Flights within K Stops	Med	Bellman-Ford k relaxation passes
1584	Min Cost to Connect All Points	Med	Kruskal or Prim MST
332	Reconstruct Itinerary	Hard	Hierholzer's Eulerian path DFS
269	Alien Dictionary	Hard	Topological sort on char DAG
1489	Find Critical and Pseudo-Critical	Hard	Tarjan's bridge algorithm
864	Shortest Path to Get All Keys	Hard	BFS with bitmask as state
1168	Optimize Water Distribution	Hard	MST with virtual well node

Phase 17 — Math & Geometry

Days 141–148 · 13 problems

Pattern: Number theory tricks · matrix rotation/spiral · fast exponentiation · slope hashmap

Week 28

LC#	Problem	Diff	Key Technique
66	Plus One	Easy	Carry simulation from last digit
202	Happy Number	Easy	Fast/slow cycle detection on digits
118	Pascal's Triangle	Easy	Each row derived from previous row
73	Set Matrix Zeroes	Med	Use first row/col as flag arrays
48	Rotate Image	Med	Transpose then reverse each row
54	Spiral Matrix	Med	Boundary shrinking iteration
179	Largest Number	Med	Custom sort by string concatenation
50	Pow(x, n)	Med	Fast exponentiation $x^{(n/2)}$ squared
43	Multiply Strings	Med	Grade-school digit-by-digit multiply
1137	N-th Tribonacci Number	Easy	Rolling 3-variable DP
2013	Detect Squares	Med	Hashmap of points, count rectangles
149	Max Points on a Line	Hard	Slope hashmap per anchor point
1232	Check If Points Are on a Line	Easy	Slope comparison from first point

Phase 18 — Bit Manipulation

Days 149–157 · 10 problems

Pattern: XOR cancels duplicates · $n \& (n-1)$ clears lowest bit · shift and build · modular bit count

Week 29

LC#	Problem	Diff	Key Technique
136	Single Number	Easy	XOR all values — duplicates cancel out
268	Missing Number	Easy	XOR or math sum $n*(n+1)/2$
191	Number of 1 Bits	Easy	$n \& (n-1)$ loop or <code>bin(n).count("1")</code>
338	Counting Bits	Easy	dp: <code>bits[i]=bits[i>>1]+(i&1)</code>
190	Reverse Bits	Easy	Shift result left, take last bit of n
371	Sum of Two Integers	Med	$a \text{ XOR } b = \text{sum}$, $(a \text{ AND } b) \ll 1 = \text{carry}$
7	Reverse Integer	Med	Mod 10 pop digits, check boundary
201	Bitwise AND of Numbers Range	Med	Find common prefix bits
137	Single Number II	Med	Count each bit position mod 3
260	Single Number III	Hard	XOR then partition by differing bit